

A Prospective Study of Long-term Intake of Dietary Fiber and Risk of Crohn's Disease and Ulcerative Colitis

Background & Aims Increased intake of dietary fiber has been proposed to reduce risk of inflammatory bowel diseases (Crohn's disease [CD], ulcerative colitis [UC]). However, few prospective studies have examined associations between long-term intake of dietary fiber and risk of incident CD or UC. **Methods** We collected and analyzed data from 170,776 women, followed over 26 y, who participated in the Nurses' Health Study, followed for 3,317,425 person-y. Dietary information was prospectively ascertained via administration of a validated semi-quantitative food frequency questionnaire every 4 y. Self-reported CD and UC were confirmed through review of medical records. Cox proportional hazards models, adjusting for potential confounders, were used to calculate hazard ratios (HRs). **Results** We confirmed 269 incident cases of CD (incidence 8/100,000 person-y) and 338 cases of UC (incidence 10/100,000 person-y). Compared to the lowest quintile of energy-adjusted cumulative average intake of dietary fiber, intake of the highest quintile (median of 24.3 g/day) was associated with a 40% reduction in risk of CD (multivariate HR for CD, 0.59; 95% confidence interval [CI], 0.39–0.90). This apparent reduction appeared to be greatest for fiber derived from fruits; fiber from cereals, whole grains, or legumes did not modify risk. In contrast, neither total intake of dietary fiber (multivariate HR, 0.82; 95% CI 0.58–1.17) nor intake of fiber from specific sources appeared to be significantly associated with risk of UC. **Conclusion** Based on data from the Nurses' Health Study, long-term intake of dietary fiber, particularly from fruit, is associated with lower risk of CD but not UC. Further studies are needed to determine the mechanisms that mediate this association.